

Código:

```
package ico.fes.aragon.inicio;

import java.util.Arrays;

public class MergeSort {
    public static int[] mergeSort(int[] data) {
        if (data.length > 1) {
            int mitad = data.length / 2;
            int[] izquierda =
Arrays.copyOfRange(data, 0, mitad);
            int[] derecha =
Arrays.copyOfRange(data, mitad, data.length);

System.out.println(Arrays.toString(izquierda) + "
--- " + Arrays.toString(derecha));

            mergeSort(izquierda);
            mergeSort(derecha);

            // Merge
            int i = 0, d = 0, k = 0;
            int[] resultado = new
int[data.length];

            while (i < izquierda.length && d <
derecha.length) {
                if (izquierda[i] < derecha[d]) {
                    resultado[k++] =
izquierda[i++];
                } else {
                    resultado[k++] =
derecha[d++];
                }
            }
        }
    }
}
```

```

        // Acomodar los restantes
        while (i < izquierda.length) {
            resultado[k++] = izquierda[i++];
        }

        while (d < derecha.length) {
            resultado[k++] = derecha[d++];
        }

        // Copiar resultado de vuelta al
        array original
        System.arraycopy(resultado, 0, data,
0, data.length);
    }

    System.out.println("regreso de rec: " +
Arrays.toString(data));
    return data;
}

    public static void main(String[] args) {
        System.out.println(".-.-.-.-.- MERGE --
.-.-.-.-.-");
        int[] info = {38, 27, 43, 3, 9, 82, 10,
19, 50, 61};

        System.out.println(Arrays.toString(mergeSort(info
)));
    }
}

```

Captures de funcionamiento

```
"C:\Program Files\Java\jdk-22\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edi
...-...-...- MERGE ...-...-...-
[38, 27, 43, 3, 9] --- [82, 10, 19, 50, 61]
[38, 27] --- [43, 3, 9]
[38] --- [27]
regreso de rec: [38]
regreso de rec: [27]
regreso de rec: [27, 38]
[43] --- [3, 9]
regreso de rec: [43]
[3] --- [9]
regreso de rec: [3]
regreso de rec: [9]
regreso de rec: [3, 9]
regreso de rec: [3, 9, 43]
regreso de rec: [3, 9, 27, 38, 43]
[82, 10] --- [19, 50, 61]
[82] --- [10]
```

```
regreso de rec: [82]
regreso de rec: [10]
regreso de rec: [10, 82]
[19] --- [50, 61]
regreso de rec: [19]
[50] --- [61]
regreso de rec: [50]
regreso de rec: [61]
regreso de rec: [50, 61]
regreso de rec: [19, 50, 61]
regreso de rec: [10, 19, 50, 61, 82]
regreso de rec: [3, 9, 10, 19, 27, 38, 43, 50, 61, 82]
[3, 9, 10, 19, 27, 38, 43, 50, 61, 82]

Process finished with exit code 0
```